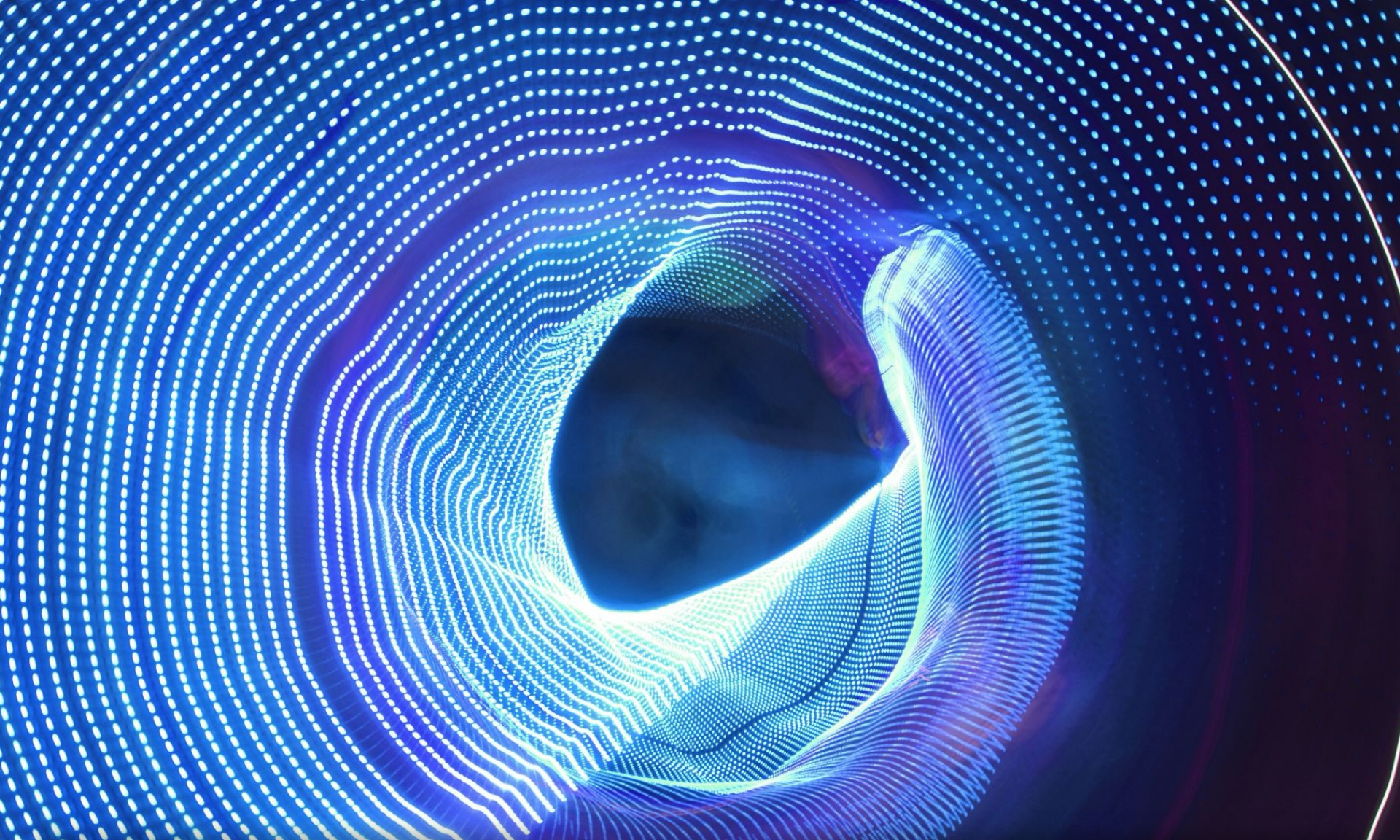
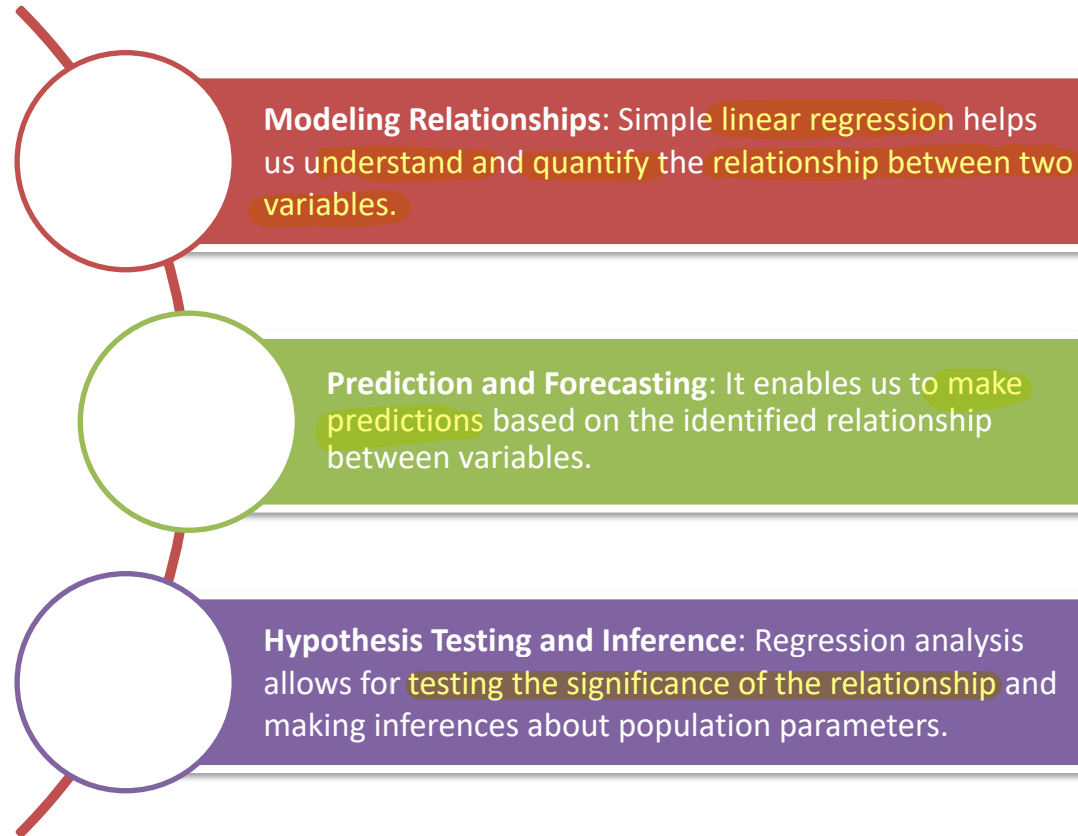




Regression

Stat 3113: Lesson 34

Motivations



Least-Squares Regression

If we have two variables that have a strong linear relationship, we would like to be able to find the equation of the line that describes the relationship. This is what regression will do for us. Least-Squares regression is one type of regression. We will see in a moment what the motivation behind it is. First, let's get some notation.

Simple Linear Regression Model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$Y_i \rightarrow i^{th}$ value of the response variable (random variable)

$X_i \rightarrow i^{th}$ value of the explanatory variable (treated as a constant)

$\beta_0 \rightarrow$ Y-intercept ...this is a parameter

$\beta_1 \rightarrow$ Slope...this is a parameter

$\varepsilon_i \rightarrow$ Random Error for the i^{th} observation (this is a random variable)

Fitted Simple Linear Regression Model:

$$\hat{y} = b_0 + b_1x$$

$\hat{y} \rightarrow$ predicted value (value the regression line predicts) of the response y at a value of x in the data set

$b_0 \rightarrow$ y-intercept of the fitted regression line...this is a statistic which estimates β_0

$b_1 \rightarrow$ slope of the regression line...this is a statistic which estimates β_1

$x \rightarrow$ explanatory variable value

$e \rightarrow$ residual or observed error ($e = y - \hat{y}$)

Now, the question is: How do we find the “best” line? In other words, how do we get the b_0 and b_1 which are the “best” estimators. One technique is called least-squares regression which minimizes the squared-error (this method is called least-squares regression). So, we will minimize:

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

We will get the values of the slope and intercept from the computer. That will give us the regression equation.

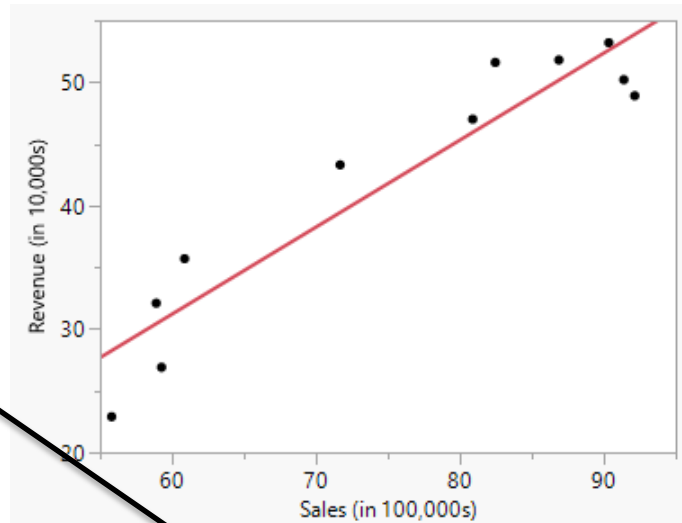
Example:

Gulf Power Company provides electricity for a large area in northwest Florida. The data gives the company's annual sales of electricity and resulting revenues for the years 1979-1989. The purpose is to see the impact sales have on revenue.

Sales (in 100,000s)	Revenue (in 10,000s)
55.8	22.9
59.27	26.9
58.9	32.1
60.87	35.7
71.68	43.3
80.89	47
86.88	51.8
82.46	51.6
90.35	53.2
91.39	50.2
92.15	48.9

OUTPUT FROM JMP:

b_0



Linear Fit

Revenue (in 10,000s) = -11.09266 + 0.7050218 * Sales (in 100,000s)

Summary of Fit

RSquare	0.885931
RSquare Adj	0.873256
Root Mean Square Error	3.876216
Mean of Response	42.14545
Observations (or Sum Wgts)	11

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	1050.2418	1050.24	69.8994
Error	9	135.2254	15.03	Prob > F
C. Total	10	1185.4673		<.0001*

Graph is called a **Line Plot**. A line plot is a scatterplot with the regression line plotted on it.

b_1

Regression Equation: $\hat{y} = -11.09 + 0.705x$

Interpretation of the Intercept and Slope

$b_0 \rightarrow$ value we expect y to be when x is 0. If $x = 0$ is in the data set, you can get this interpretation.

$b_1 \rightarrow$ amount we expect y to change when x increases by 1 unit. If slope is positive, then change will be increase; if slope is negative, will be decrease.

Example: Interpret the slope for the above example.

The regression equation is given by:

$$\hat{\text{Revenue}} = -11.09 + 0.705 \text{Sales}$$

Hence, when sales increase by 1 unit, we expect the revenue to increase by 0.705.

Hypothesis Testing:

Assumptions/Conditions:

1. Errors are normally distributed
2. Constant variance of errors
3. Independent errors

$$H_0 : \beta_1 = 0 \text{ (slope between x and y is 0)}$$

$$H_a : \beta_1 \neq 0 \text{ (slope between x and y is not 0)}$$

Analysis of Variance (ANOVA) table for simple linear regression:

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	$\sum(\hat{y} - \bar{y})^2$	SSR/DF	MSR/MSE	p-value
Residual Error	n - 2	$\sum(y - \hat{y})^2$	SSE/DF		
Total	n - 1	$\sum(y - \bar{y})^2$			

Running Hypothesis Test:

$H_0 : \beta_1 = 0$ (slope between sales and revenue is 0)

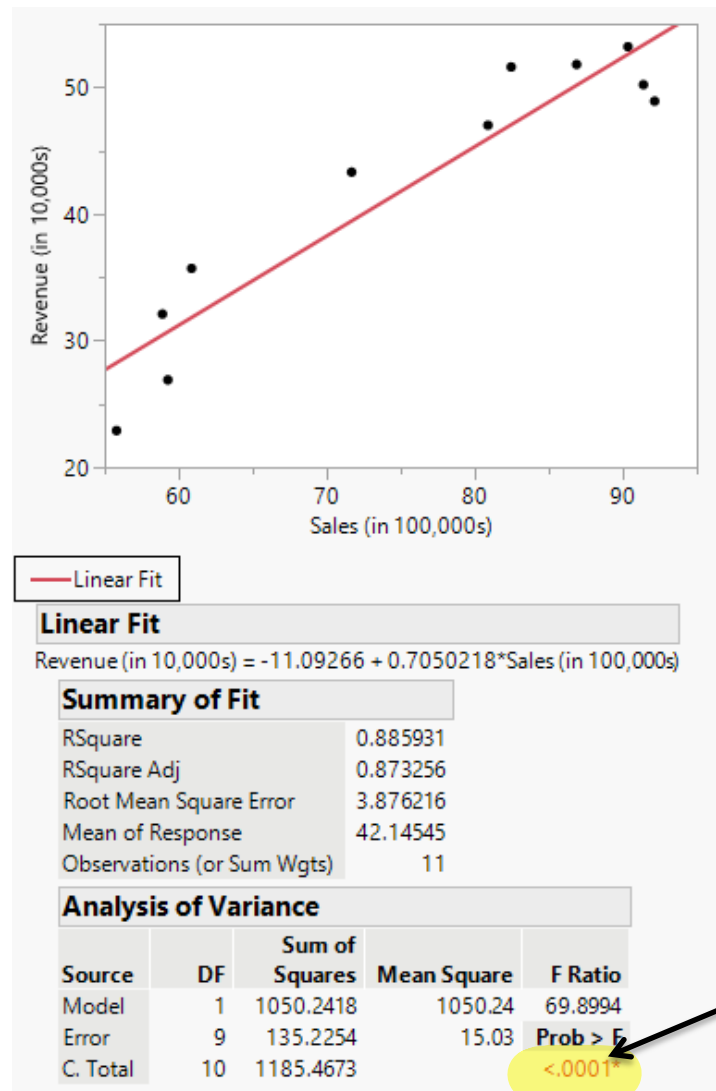
$H_a : \beta_1 \neq 0$ (slope between sales and revenue is not 0)

Two places to find the p-value:

1. ANOVA

2. Individual t-test for slope

OUTPUT FROM JMP:



Conclusion:

$\alpha=0.05$, $p\text{-value} < 0.0001$; reject H_0 . Thus, there is evidence sales and revenue have a non-zero slope.

The hypothesis test basically tests a horizontal line (slope = 0) versus a line with a non-zero slope. Thus, it won't really tell us if the line is doing a good job of prediction/explanation.

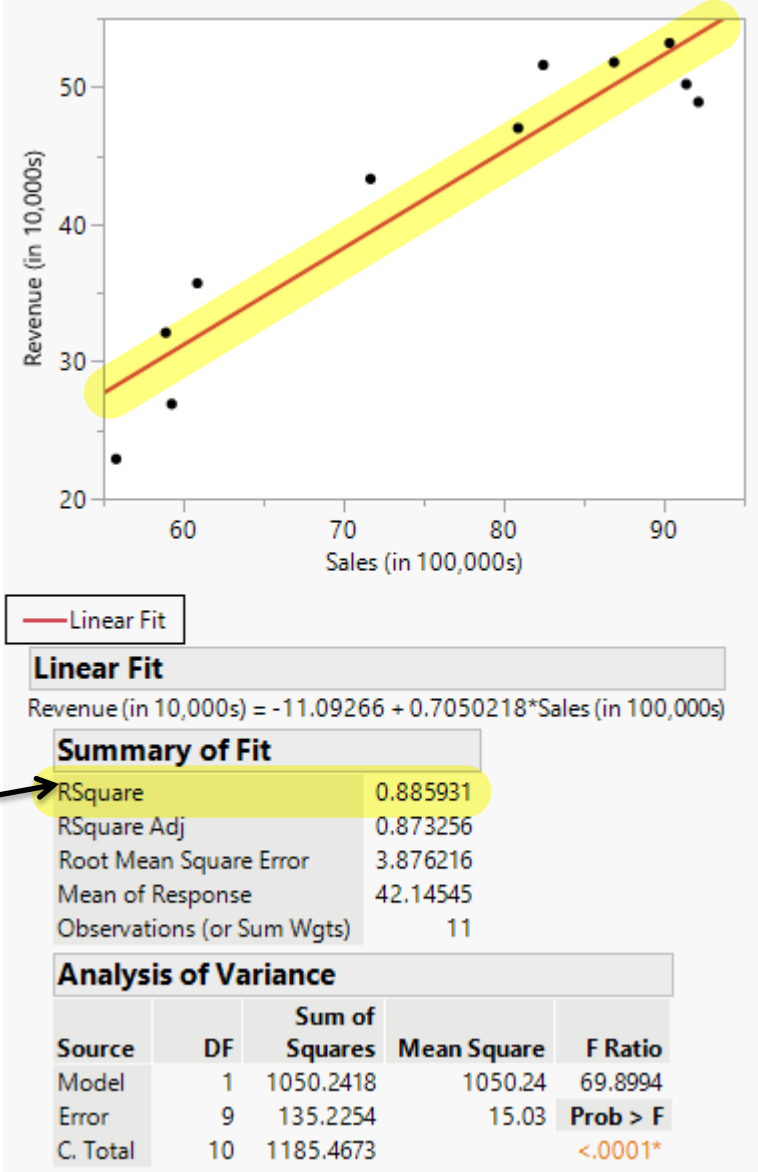
Determining if the regression line is doing a good job (r^2 value)

r^2 value → the proportion of variation in the response, y , that is explained by the line with the explanatory variable, x .

$$r^2 \cdot 100\% = \% \text{ of variation}$$

It is impossible to give a rule about r^2 and how good a line is. It really depends on the situation. It is different for each individual data analysis. Sometimes in science a value of 80%-90% or higher is what is needed. Other times, a value 70% might be good. In behavioral sciences, I've had researchers tell me a value of 35% can be wonderful. Clearly, there's no "rule".

JMP OUTPUT:



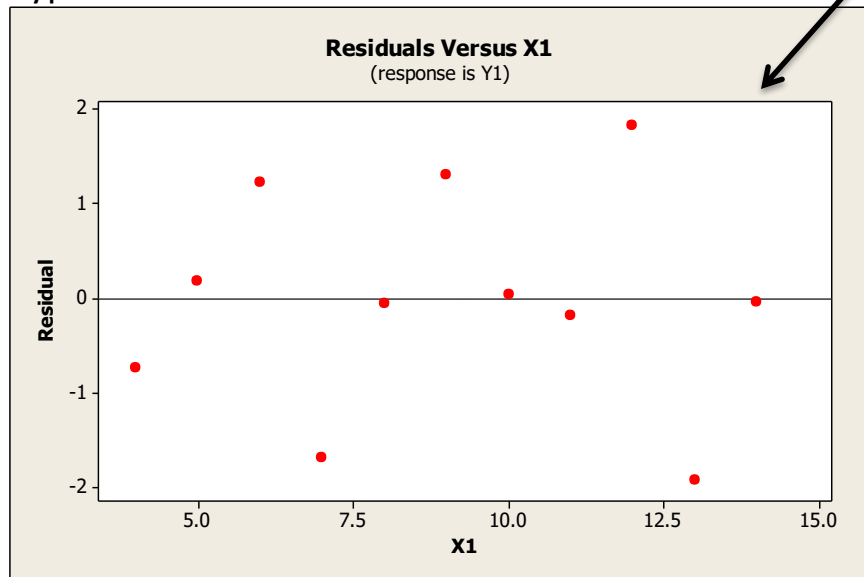
$r^2 = 88.6\%$. Thus, 88.6% of variation in Revenue has been explained by the regression line with Sales. This is a strong line.

Checking Assumptions:

1. To check normality of the errors, do a normal probability plot of the residuals
2. For rest, we will use a graph called a Residual Plot

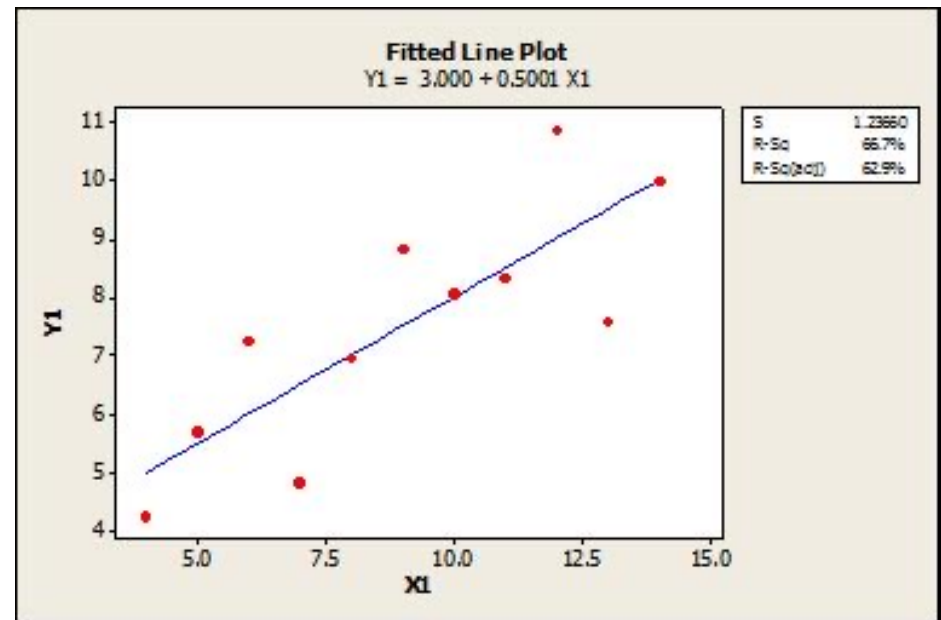
Residual is another name for the observed error ($e = y - \hat{y}$). A residual plot for a regression with one explanatory variable is a scatterplot of the residuals (y-axis) versus the explanatory variable (x-axis). A line is then drawn at $e = 0$.

Typical Residual Plot:



What we want to see is a random scattering of the points (independence) with a consistent spread (constant variance)

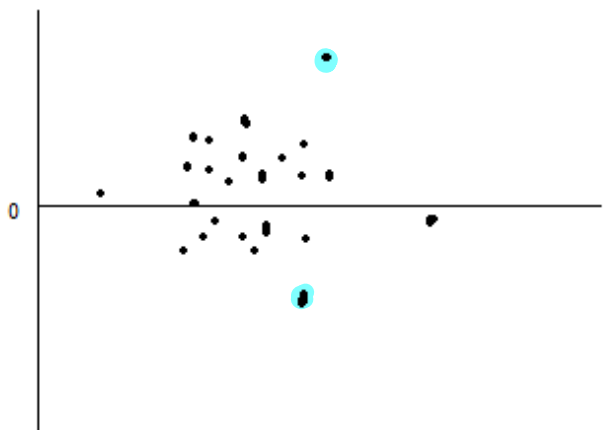
Why draw the line at $e = 0$?



Problems to look for:

1. **Outliers:** identify these first so you don't confuse them with the other problems.

Example graphs:



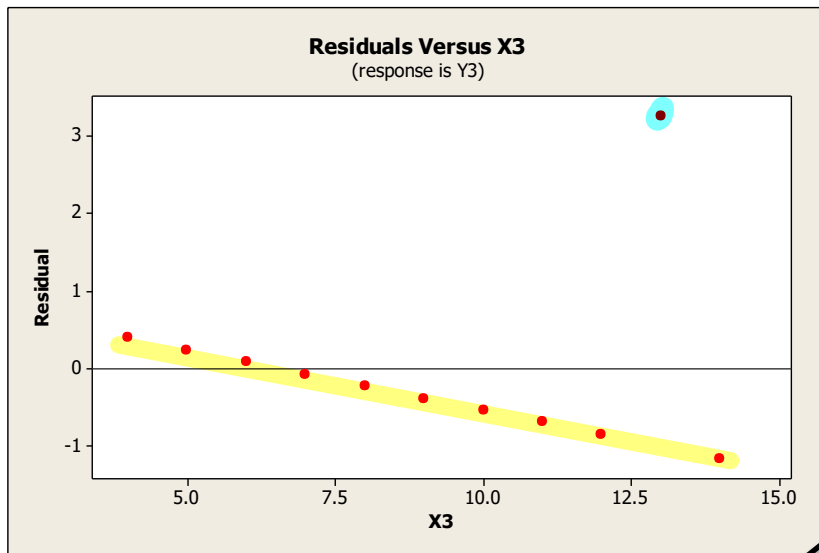
Defn: An **influential point** is an outlier which drastically affects the direction the regression line takes.

Notes:

1. Every influential point is an outlier, BUT NOT EVERY OUTLIER IS AN INFLUENTIAL POINT.
2. Influential points are often times out by themselves in the x-direction.

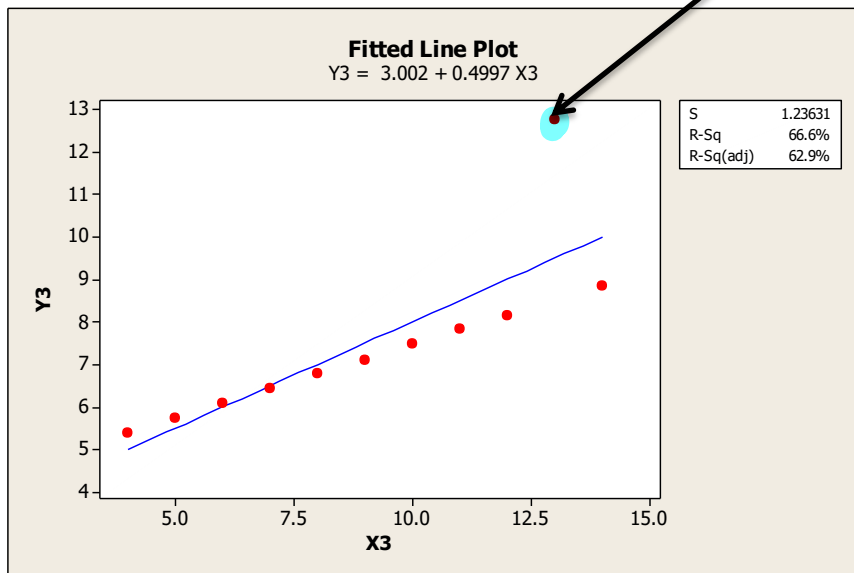
Checking for influential points:

1. Look for a point which appears to pull the regression line toward it.
2. Remove the point from step (1) from the data set.
3. Rerun the regression without that point.
4. Compare line plots (or regression equations) and determine if the line has changed its direction by a **significant** amount.



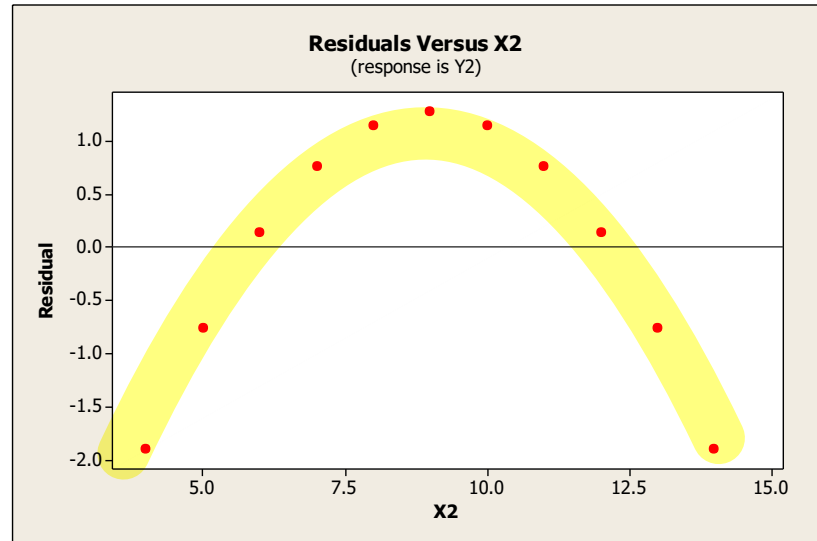
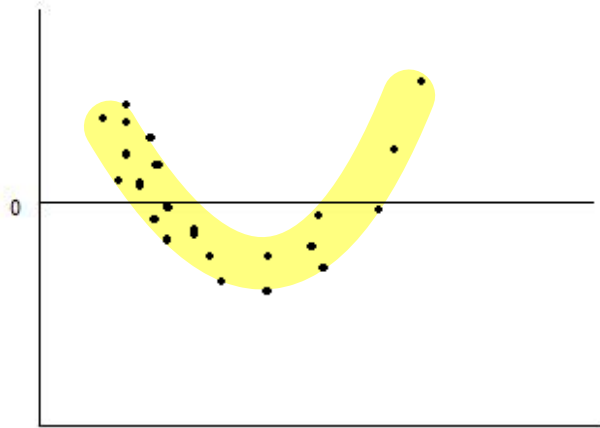
Influential point

Line Plot for above Residual Plot:



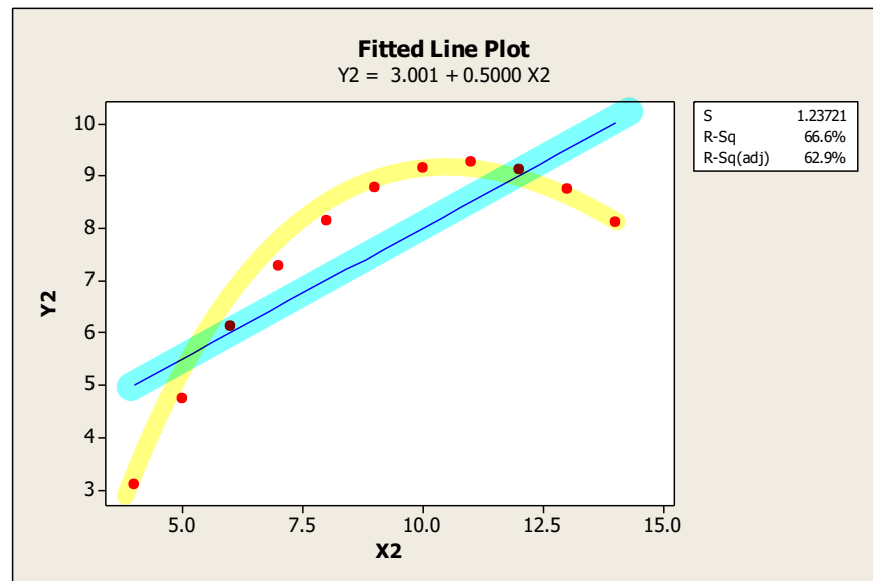
2. **Curve:** means a curve might be a better model for the data.

Example graphs:



If your line is doing a good job (high rsquared), then you will often stay with it. If not, a curve model is tried. When you see a curve in your residual plot, it means a curve model MIGHT do better than the line.

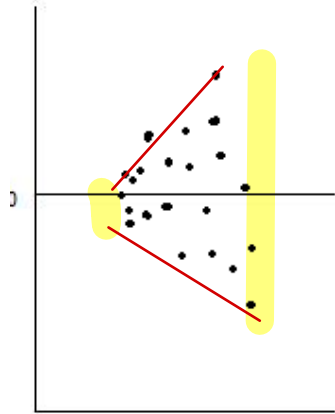
Line Plot of the above:



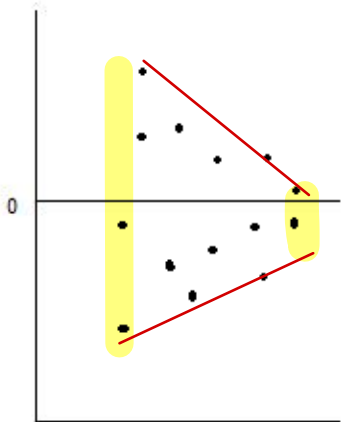
3. **Funnel**: means as the x values change the error rate is changing.

Example graphs:

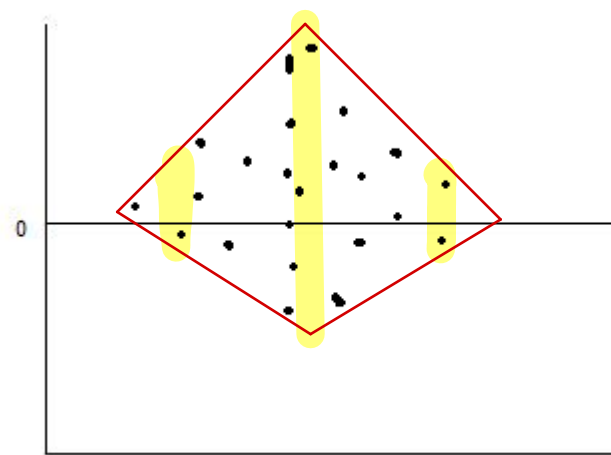
As x increases, error increases:



As x increases, error decreases:



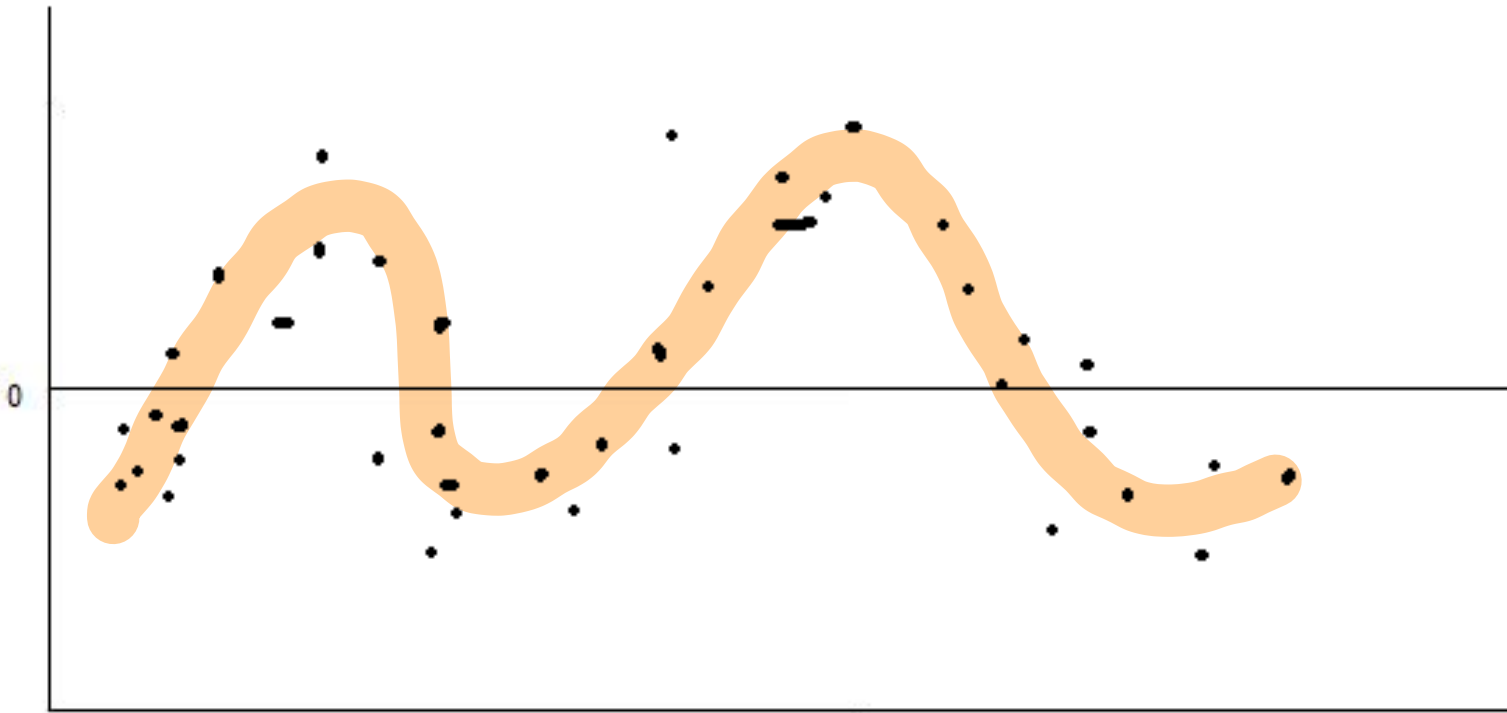
Middle x values have higher error:



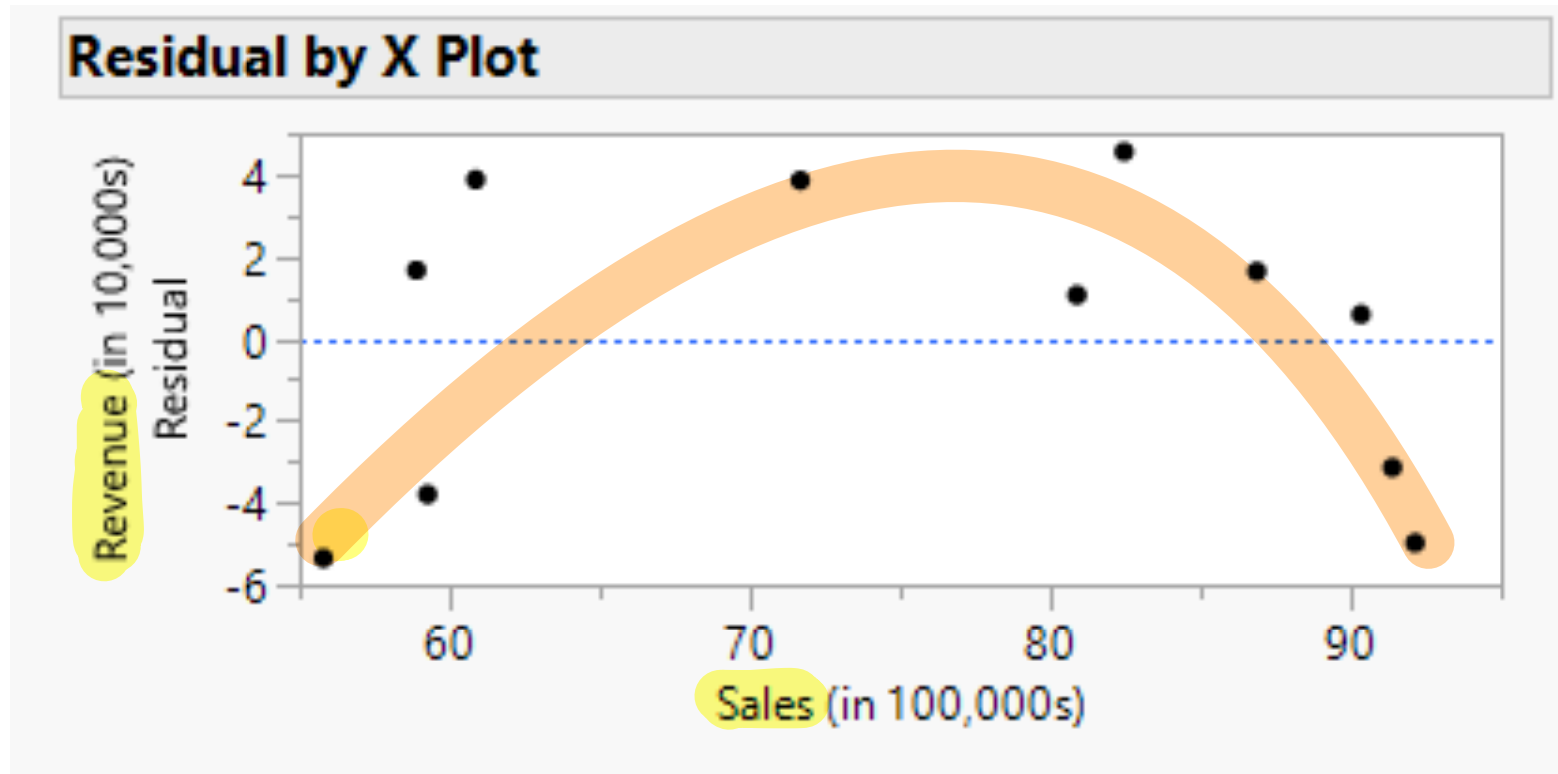
- This means the errors **don't have a constant variance**.
- There are some things which can be tried to fix it (take **natural log of y's**), but sometimes that doesn't work and a **new regression method** must be used (not **least squared**).

4. **Cyclical pattern:** means simple linear regression is not the way to go (probably a time-series)

Example graphs:

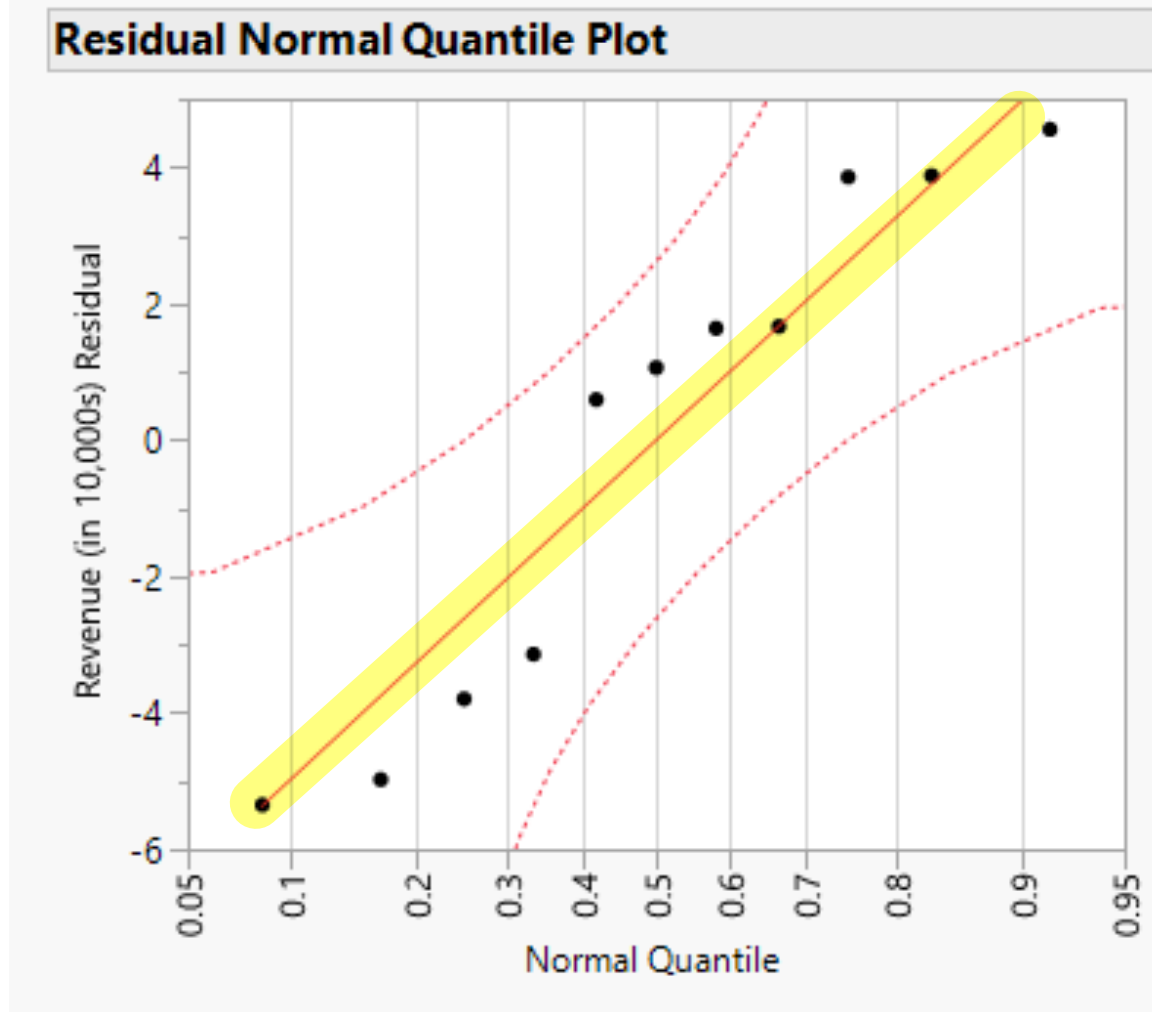


Residual Plot for our Example:



No outliers, no funnel, no cyclical pattern, but there is a curve. Curve means a curve model MIGHT be better than line.

Normal Probability Plot for our Example:



Looks fine. All data stays close to line and is in the confidence bands. Residuals appear to come from approximate normal distribution.

Cautions in Correlation and Regression Analysis

1. **Extrapolation:** When a value of the explanatory variable OUTSIDE THE ORIGINAL DATA X-VALUE RANGE is plugged into a regression equation to get a predicted value. This is a BAD thing to do.

Note: Many times (BUT NOT ALWAYS) extrapolating will give predicted values that don't make much sense, but the real problem is predicted for a value not present within the original data set range.

2. **Association Does NOT Imply Causation:** Just because two variables are strongly related to one another, does NOT mean x causes y to change.

Example: Age and Salary; Ice Cream Sales and Crime Rate

3. Lurking Variable: A variable not in the study which strongly affects the relationship between the response and explanatory variables.

Example: In the examples before, what are the lurking variables?

For example, during periods of economic downturn, overall electricity consumption may decrease, affecting sales and revenues.

4. Gaps in the data: Look for gaps in the scatterplot/line plot to indicate potential problems.

WARNING: A causal relationship is VERY difficult to get using statistics. In order for it to even be discussed, a well-designed and implemented experiment must have been done to collect the data.



Reading

- Chapter 2 – Sections 2.2, 2.3
- Chapter 8 – Sections 8.1, 8.2

Next Topic



Chapter 2 – Sections 2.2, 2.3

Chapter 8 – Sections 8.1, 8.2



Regression Examples